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HW 2
CHE 321

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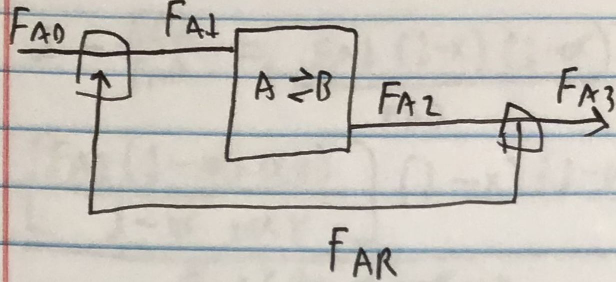


Mass balances

$F_{A1} = F_{A0} + F_{AR}$

$F_{A2} = F_{A1}$

$F_{AR} = F_{A2} - F_{A3}$

We want $X_{tot}(k_1, \dot{V}_0, \dot{V}_R, V, n)$

$X_{tot} = 1 - \frac{F_{A3}}{F_{A0}}$ split fraction $\alpha = \frac{F_{AR}}{F_{A2}}$

$\frac{F_{AR}}{F_{A2}} = \frac{C_{AR}\dot{V}_R}{C_{A2}\dot{V}_2} = \frac{\dot{V}_R}{n\dot{V}_2}$ $V_2 = \dot{V}_0 + \dot{V}_R$ $\Rightarrow \frac{\dot{V}_R}{n(\dot{V}_0+\dot{V}_R)}$

$F_{A1} = F_{A0} + \alpha F_{A2} \Rightarrow F_{A0} + \underbrace{\frac{\dot{V}_R}{n(\dot{V}_0+\dot{V}_R)}}_{\alpha} F_{A2}$ $F_{A2} = F_{A1}(1-X)$

$F_{A1} = F_{A0} + \frac{\dot{V}_R}{n(\dot{V}_0+\dot{V}_R)}[F_{A1} - F_{A1}X]$

$F_{A1} = F_{A0} + \alpha \underbrace{F_{A1} - \alpha F_{A1}X}_{F_{A1}(1-X)} \Rightarrow F_{A0} = F_{A1}(1 + \alpha X - \alpha)$

$F_{A1} = F_{A0}/(1 + \alpha X - \alpha)$

Ballance in CSTR

$F_{A1} - F_{A2} + (-k_f C_{A2} + k_f C_{B2})V = 0$ $C_{A2} = C_{A1}(1-X)$

$F_{A1} - [F_{A1}(1-X)] + k_f V((C_{A1}(1-X) + C_{A1}X) = 0$ $C_{B2} = C_{A1}X$

$\cancel{F_{A1}} - \cancel{F_{A1}} + F_{A1}X + k_f V((C_{A1} + 2C_{A1}X) = 0$

$F_{A1}X + k_f V C_{A1}(1+2X) = 0$ $C_{A1} = \frac{F_{A1}}{\dot{V}_1} = \frac{F_{A1}}{\dot{V}_0+\dot{V}_R}$

$F_{A1}X + \frac{k_f V F_{A1}}{\dot{V}_0+\dot{V}_R}(1+2X) = 0$

$F_{A1}\left[X + \frac{k_f V(1-2X)}{\dot{V}_0+\dot{V}_R}\right] = 0$

$X + \frac{k_f V}{\dot{V}_0+\dot{V}_R} - \frac{2k_f V X}{\dot{V}_0+\dot{V}_R} = 0$ $X - 2X\frac{V k_f}{\dot{V}_0+\dot{V}_R} = -\frac{k_f V}{\dot{V}_0+\dot{V}_R}$

$X\left(1 - \frac{2Vk_f}{\dot{V}_0+\dot{V}_R}\right) = \frac{-k_f V}{\dot{V}_0+\dot{V}_R}$ $X = \frac{-k_f V/\dot{V}_0+\dot{V}_R}{1-(2Vk_f/\dot{V}_0+\dot{V}_R)}$

$F_{A3} = F_{A2} - F_{AR} = F_{A2} - \alpha F_{AR} \Rightarrow F_{A3} = F_{A2}(1-\alpha) = F_{A1}(1-X)(1-\alpha) = F_{A0}$

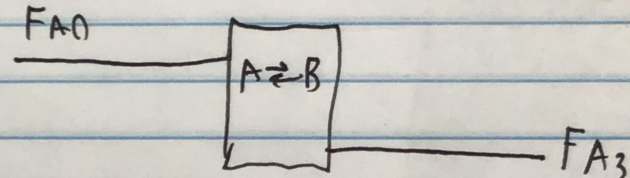
~~FA3 = FA0 from overall balance~~

$$1 - \frac{F_{A3}}{F_{A0}} = 1 - \frac{F_{A1}(1-x)(1-\alpha)}{F_{A0}} \qquad \frac{[F_{A0}/(1+\alpha x-\alpha)](1-x)(1-\alpha)}{F_{A0}}$$

$$\frac{\left[\frac{F_{A1}(1-\alpha+\alpha x)}{1-\alpha+\alpha x}\right](1-x)(1-\alpha)}{F_{A1}(1-\alpha+\alpha x)} = \frac{F_{A1}(1-x)(1-\alpha)}{F_{A1}(1-\alpha+\alpha x)}$$

$$\boxed{\Rightarrow X_{tot} = 1-\frac{F_{A3}}{F_{A0}} = 1-\frac{1-x-\alpha+\alpha x}{1-\alpha+\alpha x} \quad \text{where } \alpha = \frac{\dot v_R}{n(\dot v_0+\dot v_R)} \quad \& \quad x = \frac{-k_f V/\dot v_0+\dot v_R}{1-(2Vk_f/\dot v_0+\dot v_R)}}$$

b No recycle



$X_{tot} = 1-\frac{F_{A3}}{F_{A0}}$ $\qquad F_{A3}=F_{A0}(1-x)$

Inside Balar

$F_{A0}-F_{A3}+(-k_f C_{A3}+k_f C_{B3})V=0 \qquad C_{A3}=C_{A0}(1-x)$

$F_{A0}-[F_{A0}(1-x)]-k_f[C_{A0}(1-x)+k_f C_{A0}x]V=0 \qquad C_{B3}=C_{A0}x$

$F_{A0}x-[k_f C_{A0}+k_f C_{A0}x+k_f C_{A0}x]V=0$

$F_{A0}x+k_f C_{A0}V[(-1+2x)]=0 \qquad C_{A0}=\frac{F_{A0}}{\dot v_0}$

$F_{A0}x+k_f\frac{F_{A0}V}{\dot v_0}[2x-1]=0$

$F_{A0}\left[x+\frac{2k_f V x}{\dot v_0}\right]-\frac{k_f V}{\dot v_0}=0$

$x+\frac{2k_fVx}{\dot v_0}=\frac{k_fV}{\dot v_0} \qquad x=\frac{(k_fV/\dot v_0)}{[(1+2k_f(V/\dot v_0)]}$

$$\boxed{x = 1-\frac{F_{A3}}{F_{A0}} = \frac{F_{A0}(1-x)}{F_{A0}} \quad X_{tot}=1-x \quad \text{where } x=\frac{(k_fV/\dot v_0)}{1+(2k_fV)/\dot v_0}}$$